mysterious, baffling, impossible to understand. That message goes against the basic principles that science stands for, which include the idea that the world is fundamentally intelligible. We have something of a mental block when it comes to quantum mechanics, and we need a bit of quantum therapy to help get past it.

When we teach quantum mechanics to students, they are taught a list of rules. Some of the rules are of a familiar type: there's a mathematical description of quantum systems, plus an explanation of how such systems evolve over time. But then there are a bunch of extra rules that have no analogue in any other theory of physics. These extra rules tell us what happens when we *observe* a quantum system, and that behavior is completely different from how the system behaves when we're not observing it. What in the world is going on with that?

There are basically two options. One is that the story we've been telling our students is woefully incomplete, and in order for quantum mechanics to qualify as a sensible theory we need to understand what a "measurement" or "observation" is, and why it seems so different from what the system does otherwise. The other option is that quantum mechanics represents a violent break from the way we have always thought about physics before, shifting from a view where the world exists objectively and independently of how we perceive it, to one where the act of observation is somehow fundamental to the nature of reality.

In either case, the textbooks should by all rights spend time exploring these options, and admit that even though quantum mechanics is extremely successful, we can't claim to be finished developing it just yet. They don't. For the most part, they pass over this issue in silence, preferring to stay in the physicist's comfort zone of writing down equations and challenging students to solve them.

That's embarrassing. And it gets worse.

You might think, given this situation, that the quest to understand quantum mechanics would be the single biggest goal in all of physics. Millions of dollars of grant money would flow to researchers in quantum foundations, the brightest minds would flock to the problem, and the most